# Policy Catalog Azure Permissions List

This document contains a list of cloud provider permissions required by each Azure policy template in the Flexera Cloud Cost Optimization Policy Catalog.

- Required permissions may change as policy templates in the catalog are updated. An up-to-date version of this PDF file is available in the Flexera Policy Template Github Repository.
- *Required* permissions are required for the policy template in question to work correctly; *optional* permissions enable additional functionality.
- Permissions with a green *Read Only* label are read-only, while those with a red *Write* label can potentially make changes to your cloud environment.

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## AKS Node Pools Without Autoscaling 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.ContainerService/managedClusters/agentPools/read *Read Only*
Microsoft.ContainerService/managedClusters/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## AKS Node Pools Without Zero Autoscaling 3.4.3

**Azure Resource Manager Required Permissions:**
Microsoft.ContainerService/managedClusters/agentPools/read *Read Only*
Microsoft.ContainerService/managedClusters/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure AHUB Utilization with Manual Entry 4.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/locations/vmSizes/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Advisor Carbon Reduction Recommendations 0.4.3

**Azure Resource Manager Required Permissions:**
Microsoft.Advisor/recommendations/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Advisor Compute Instances Recommendations 0.5.2

**Azure Resource Manager Required Permissions:**
Microsoft.Advisor/recommendations/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Blob Storage Accounts Without Logging Enabled 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**

Storage Blob Data Reader *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Blob Storage Accounts Without Soft Delete Enabled 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/blobServices/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Blob Storage Optimization 4.2.5

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read *Read Only*
Microsoft.Storage/storageAccounts/blobServices/containers/list *Read Only*

**Azure Storage Optional Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete *Write*
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write *Write*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Bring-Your-Own-License (BYOL) Report 0.5.2

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Cheaper Regions 1.2.3

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

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## Azure Compute Instances Time Powered Off Report 0.5.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Cost Report - VM Instance Cost Per Hour 0.1.3

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

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## Azure Credential 0.3.2

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Data Lake Optimization 0.4.5

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**

Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read *Read Only*
Microsoft.Storage/storageAccounts/blobServices/containers/list *Read Only*

**Azure Storage Optional Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/delete *Write*
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Databricks Rightsize Compute Instances 0.8.3

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/skus/read *Read Only*
Microsoft.Databricks/workspaces/read *Read Only*
Microsoft.Insights/*/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Deprecated Storage Accounts 0.4.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Disallowed Regions 4.4.2

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure End-of-Life Resources 0.2.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Expiring Certificates 4.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Web/certificates/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Web/certificates/delete *Write*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Expiring Reserved Instances 3.1.4

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

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## Azure Expiring Savings Plans 0.3.0

**Azure Resource Manager Required Permissions:**

Microsoft.Billing/billingAccounts/billingProfiles/read *Read Only*
Microsoft.Billing/billingAccounts/read *Read Only*
Microsoft.BillingBenefits/savingsPlanOrders/read *Read Only*
Microsoft.BillingBenefits/savingsPlanOrders/savingsPlans/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

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## Azure Guest Users 3.1.2

**Microsoft Graph Required Permissions:**
User.Read.All *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

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## Azure Hybrid Use Benefit for Linux Server 5.6.3

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Hybrid Use Benefit for SQL 4.9.3

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/managedInstances/read *Read Only*
Microsoft.Sql/servers/databases/read *Read Only*
Microsoft.Sql/servers/elasticPools/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.SqlVirtualMachine/sqlVirtualMachines/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Sql/managedInstances/write *Write*
Microsoft.Sql/servers/databases/write *Write*
Microsoft.Sql/servers/elasticPools/write *Write*

Microsoft.SqlVirtualMachine/sqlVirtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Hybrid Use Benefit for Windows Server 5.7.3

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Idle ML Online Endpoints 0.2.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.MachineLearningServices/workspaces/onlineEndpoints/deployments/read *Read Only*
Microsoft.MachineLearningServices/workspaces/onlineEndpoints/read *Read Only*
Microsoft.MachineLearningServices/workspaces/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.MachineLearningServices/workspaces/onlineEndpoints/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Instances not running FlexNet Inventory Agent 4.3.5

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Long Running Instances 6.4.5

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Long Stopped Compute Instances 6.4.1

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/disks/read *Read Only*
Microsoft.Compute/virtualMachines/instanceView/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Insights/eventtypes/management/values/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Missing Subscriptions 0.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

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## Azure MySQL Flexible Servers Without Secure TLS 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBforMySQL/flexibleServers/configurations/read *Read Only*
Microsoft.DBforMySQL/flexibleServers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure MySQL Servers Without Enforced SSL 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBforMySQL/servers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Network Security Groups With Inbound RDP Open 3.3.2

**Azure Resource Manager Required Permissions:**
Microsoft.Network/networkSecurityGroups/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Network Security Groups With Inbound SSH Open 3.3.2

**Azure Resource Manager Required Permissions:**
Microsoft.Network/networkSecurityGroups/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure New Marketplace Products 1.0.0

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Old Snapshots 7.6.1

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/snapshots/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/snapshots/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Overutilized Compute Instances 0.4.5

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/locations/vmSizes/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Policy Audit 3.2.2

**Azure Resource Manager Required Permissions:**
Microsoft.Authorization/policyAssignments/read *Read Only*
Microsoft.Authorization/policyDefinitions/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure PostgreSQL Servers With Bad Log Settings 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBForPostgreSql/servers/configurations/read *Read Only*
Microsoft.DBForPostgreSql/servers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure PostgreSQL Servers With Insufficient Log Retention 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBForPostgreSql/servers/configurations/read *Read Only*
Microsoft.DBForPostgreSql/servers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure PostgreSQL Servers Without Connection Throttling 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBForPostgreSql/servers/configurations/read *Read Only*
Microsoft.DBForPostgreSql/servers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure PostgreSQL Servers Without Infrastructure Encryption 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.DBForPostgreSql/servers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**

billing_center_viewer *Read Only*

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## Azure Publicly-Accessible Blob Containers 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/list *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Publicly-Accessible SQL Managed Instances 3.3.5

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/managedInstances/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Sql/managedInstances/delete *Write*
Microsoft.Sql/managedInstances/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Queue Storage Accounts Without Logging Enabled 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**
Storage Queue Data Reader *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Regulatory Compliance 3.1.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*

Microsoft.Security/regulatoryComplianceStandards/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Reserved Instances Recommendations 4.9.2

**Azure Resource Manager Required Permissions:**
Microsoft.Consumption/reservationRecommendations/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/subscriptions/resourceGroups/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Billing/billingAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Reserved Instances Utilization 4.1.3

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Rightsize Compute Instances 6.7.2

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/locations/vmSizes/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Insights/*/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize Compute Instances (Cross-Family) 0.3.2

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/skus/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Rightsize Managed Disks 3.0.2

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/disks/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize MySQL Flexible Servers 1.0.0

**Azure Resource Manager Required Permissions:**
Microsoft.DBforMySQL/flexibleServers/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.DBforMySQL/flexibleServers/delete *Write*
Microsoft.DBforMySQL/flexibleServers/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize MySQL Single Servers 0.6.3

**Azure Resource Manager Required Permissions:**
Microsoft.DBforMySQL/servers/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.DBforMySQL/servers/delete *Write*
Microsoft.DBforMySQL/servers/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize NetApp Resources 2.4.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.NetApp/netAppAccounts/capacityPools/read *Read Only*
Microsoft.NetApp/netAppAccounts/capacityPools/volumes/read *Read Only*
Microsoft.NetApp/netAppAccounts/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.NetApp/netAppAccounts/capacityPools/volumes/write *Write*
Microsoft.NetApp/netAppAccounts/capacityPools/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize PostgreSQL Flexible Servers 0.2.3

**Azure Resource Manager Required Permissions:**
Microsoft.DBforPostgreSQL/flexibleServers/read *Read Only*

Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.DBforPostgreSQL/flexibleServers/delete *Write*
Microsoft.DBforPostgreSQL/flexibleServers/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize PostgreSQL Single Servers 0.2.3

**Azure Resource Manager Required Permissions:**
Microsoft.DBforPostgreSQL/servers/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.DBforPostgreSQL/servers/delete *Write*
Microsoft.DBforPostgreSQL/servers/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize SQL Database Storage 0.5.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/locations/capabilities/read *Read Only*
Microsoft.Sql/servers/databases/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize SQL Databases 7.0.0

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/databases/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Sql/servers/databases/delete *Write*
Microsoft.Sql/servers/databases/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize SQL Managed Instance Storage 0.6.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/managedInstances/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Sql/managedInstances/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize SQL Managed Instances 0.7.3

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/managedInstances/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Sql/managedInstances/delete *Write*
Microsoft.Sql/managedInstances/write *Write*

**Flexera Required Permissions:**

billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rightsize Synapse SQL Pools 0.5.3

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Synapse/workspaces/read *Read Only*
Microsoft.Synapse/workspaces/sqlPools/pause/action *Read Only*
Microsoft.Synapse/workspaces/sqlPools/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Rule-Based Dimension For Tenant ID 0.1.3

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
ca_user *Read Only*
policy_viewer *Read Only*
rule_based_dimensions_manager *Read Only*

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## Azure Rule-Based Dimension From Resource Group Tags 2.2.2

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/subscriptions/resourceGroups/read *Read Only*
Microsoft.Resources/tags/read *Read Only*

**Flexera Required Permissions:**
observer *Read Only*
policy_viewer *Read Only*
rule_based_dimensions_manager *Read Only*

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## Azure Rule-Based Dimension From Subscription Tags 2.2.2

**Azure Resource Manager Required Permissions:**

Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/tags/read *Read Only*

**Flexera Required Permissions:**
observer *Read Only*
policy_viewer *Read Only*
rule_based_dimensions_manager *Read Only*

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## Azure SQL Databases Without Encryption 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/databases/read *Read Only*
Microsoft.Sql/servers/databases/transparentDataEncryption/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure SQL Servers Vulnerability Assessment Does Not Notify Admins 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.Sql/servers/vulnerabilityAssessments/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

---

## Azure SQL Servers Vulnerability Assessment Without Email Notifications 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.Sql/servers/vulnerabilityAssessments/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure SQL Servers Vulnerability Assessment Without Periodic Scans 3.3.4

**Azure Resource Manager Required Permissions:**

Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.Sql/servers/vulnerabilityAssessments/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

---

## Azure SQL Servers With Insufficient Auditing Retention 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/auditingSettings/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure SQL Servers Without Active Directory Admin 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/administrators/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure SQL Servers Without Advanced Threat Protection (ATP) 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.Sql/servers/securityAlertPolicies/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure SQL Servers Without Auditing Enabled 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/auditingSettings/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

## Azure SQL Servers Without Elastic Pools 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/elasticPools/read *Read Only*
Microsoft.Sql/servers/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure SQL Servers Without Vulnerability Assessment (VA) Enabled 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Sql/servers/read *Read Only*
Microsoft.Sql/servers/vulnerabilityAssessments/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Savings Plan Recommendations 3.8.2

**Azure Resource Manager Required Permissions:**
Microsoft.CostManagement/benefitRecommendations/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/subscriptions/resourceGroups/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Savings Plan Utilization 0.2.3

**Azure Resource Manager Required Permissions:**
Microsoft.Billing/billingAccounts/billingProfiles/read *Read Only*
Microsoft.Billing/billingAccounts/read *Read Only*
Microsoft.BillingBenefits/savingsPlanOrders/read *Read Only*
Microsoft.BillingBenefits/savingsPlanOrders/savingsPlans/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

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## Azure Savings Realized from Reservations 3.9.11

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

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## Azure Schedule Instance 7.2.5

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/deallocate/action *Write*
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/instanceView/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Compute/virtualMachines/start/action *Write*
Microsoft.Compute/virtualMachines/write *Write*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/tenants/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Sentinel Commitment Tier Recommendations 0.5.2

**Azure Resource Manager Required Permissions:**
Microsoft.OperationalInsights/workspaces/query/action *Read Only*
Microsoft.OperationalInsights/workspaces/read *Read Only*
Microsoft.OperationsManagement/solutions/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.SecurityInsights/onboardingStates/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Storage Accounts Allowing Default Network Access 3.4.2

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Storage Accounts Without Secure TLS 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

---

## Azure Storage Accounts Without Secure Transfer 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Storage Accounts Without Trusted Microsoft Services Access 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Storage Accounts without Lifecycle Management Policies 4.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/managementPolicies/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Subscriptions Without High Severity Alerts 3.3.1

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Security/securityContacts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Subscriptions Without Log Analytics Auto-Provisioning 3.3.1

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Security/autoProvisioningSettings/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Subscriptions Without Owner Security Alerts 3.2.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Security/securityContacts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure Subscriptions Without Security Contact Email 3.2.2

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Security/securityContacts/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

## Azure Superseded Compute Instances 4.4.3

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/locations/vmSizes/read *Read Only*
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Table Storage Accounts Without Logging Enabled 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Storage Required Permissions:**
Storage Table Data Reader *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Tag Cardinality Report 3.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/subscriptions/resourceGroups/read *Read Only*
Microsoft.Resources/subscriptions/resources/read *Read Only*
Microsoft.Resources/tags/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Untagged Resources 5.0.4

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Resources/subscriptions/resourceGroups/read *Read Only*
Microsoft.Resources/subscriptions/resources/read *Read Only*
Microsoft.Resources/tags/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Resources/tags/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Untagged Virtual Machines 2.0.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/virtualMachines/delete *Write*
Microsoft.Compute/virtualMachines/powerOff/action *Write*
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused App Service Plans 0.5.2

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Web/serverfarms/read *Read Only*
Microsoft.Web/serverfarms/sites/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Web/serverfarms/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused Firewalls 0.6.1

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Network/azureFirewalls/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Network/azureFirewalls/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused IP Addresses 7.7.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/eventtypes/management/values/read *Read Only*
Microsoft.Network/publicIPAddresses/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Network/publicIPAddresses/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused Load Balancers 0.5.1

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/metrics/read *Read Only*
Microsoft.Network/loadBalancers/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**

Microsoft.Network/loadBalancers/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused Storage Accounts 0.5.2

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/*/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Storage/storageAccounts/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Storage/storageAccounts/delete *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused Virtual Network Gateways 0.4.3

**Azure Resource Manager Required Permissions:**
Microsoft.Network/virtualNetworkGateways/connections/read *Read Only*
Microsoft.Network/virtualNetworkGateways/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Network/virtualNetworkGateways/delete *Write*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Unused Volumes 8.6.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/disks/read *Read Only*
Microsoft.Insights/metrics/read *Read Only*

Microsoft.Resources/subscriptions/read *Read Only*

**Azure Resource Manager Optional Permissions:**
Microsoft.Compute/disks/delete *Write*
Microsoft.Compute/snapshots/read *Write*
Microsoft.Compute/snapshots/write *Write*
Microsoft.Compute/virtualMachines/read *Write*
Microsoft.Compute/virtualMachines/write *Write*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Usage Report - Instance Time Used 1.0.14

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Azure VMs Not Using Managed Disks 4.3.4

**Azure Resource Manager Required Permissions:**
Microsoft.Compute/virtualMachines/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

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## Azure Web Apps With Unoptimized Scaling 0.5.3

**Azure Resource Manager Required Permissions:**
Microsoft.Insights/autoscalesettings/read *Read Only*
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Web/sites/read *Read Only*

**Flexera Required Permissions:**
policy_viewer *Read Only*

**Flexera Optional Permissions:**
policy_manager *Read Only*

---

## Azure Web Apps Without Secure TLS 3.3.3

**Azure Resource Manager Required Permissions:**
Microsoft.Resources/subscriptions/read *Read Only*
Microsoft.Web/sites/config/read *Read Only*
Microsoft.Web/sites/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*

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## Common Bill Ingestion from Azure Blob Storage 0.2.10

**Azure Storage Required Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
csm_bill_upload_admin *Read Only*
policy_viewer *Read Only*

---

## Common Bill Ingestion from Azure Blob Storage (Iterating) 1.0.0

**Azure Storage Required Permissions:**
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/list/action *Read Only*
Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read *Read Only*
Microsoft.Storage/storageAccounts/blobServices/containers/read *Read Only*

**Flexera Required Permissions:**
billing_center_viewer *Read Only*
csm_bill_upload_admin *Read Only*

**Flexera Optional Permissions:**
org_owner *Read Only*

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